

NGT resource allocation

The NGT GPU allocation problem, measured 30 days of cluster telemetry, and what a policy would change

Felice Pantaleo (CERN)
July 2026

Method in one slide

Data

Pod lifecycle, requests, placement, cordons and per-GPU utilization from cluster monitoring, 30 days at 5 min resolution.

Scope

1571 GPU/MIG requests from 114 users on the on-premise pools. STEAM Academy accounts and cloud T4 nodes excluded.

Attribution

Users mapped to WP1 to WP4 via the project roster, department rules and manual classification: 99.4% of GPU-hours attributed.

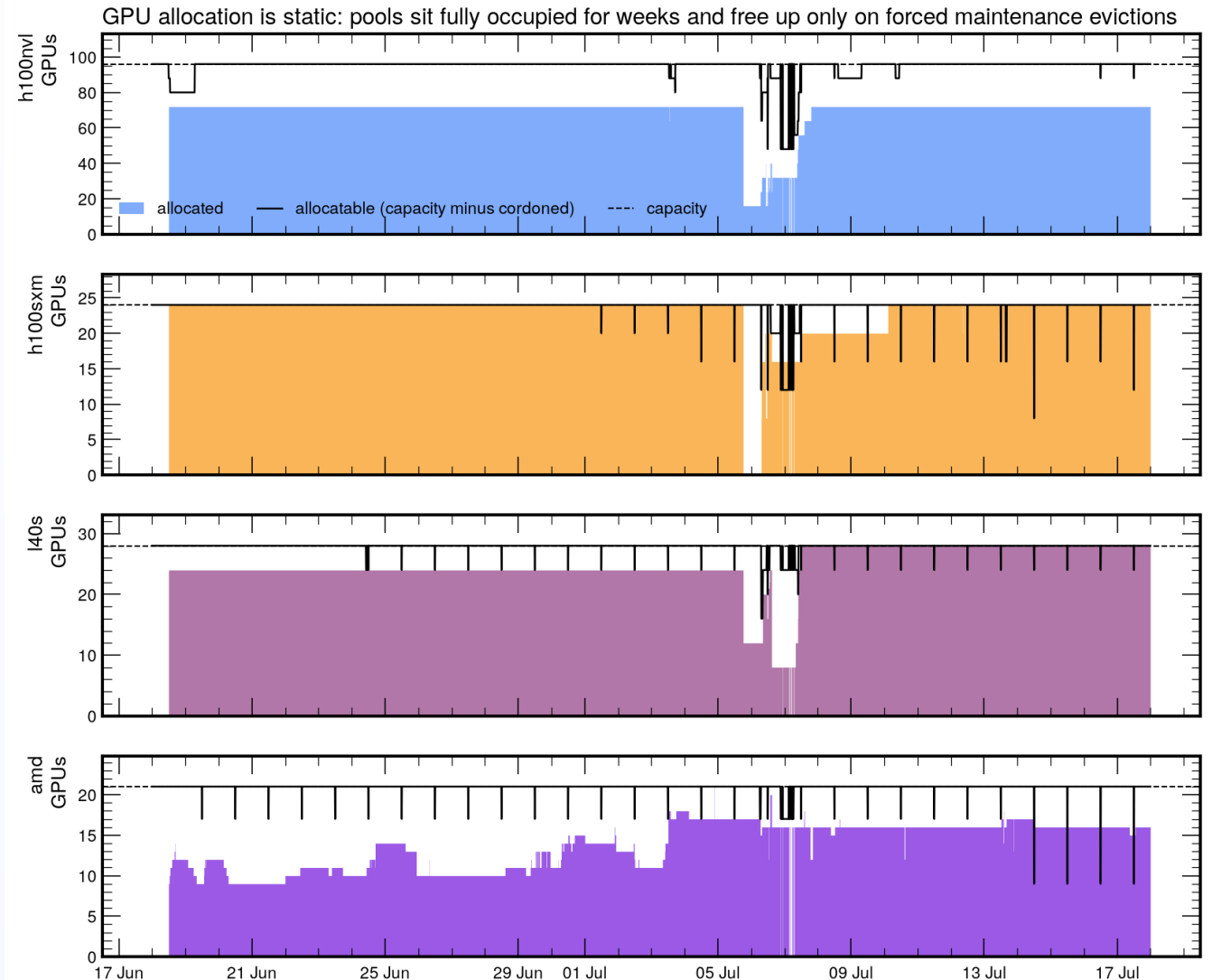
A request = one pod asking for GPUs. Wait = creation to scheduling.

Idle = GPU utilization below 5%.

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Allocation is static: pools sit full for weeks

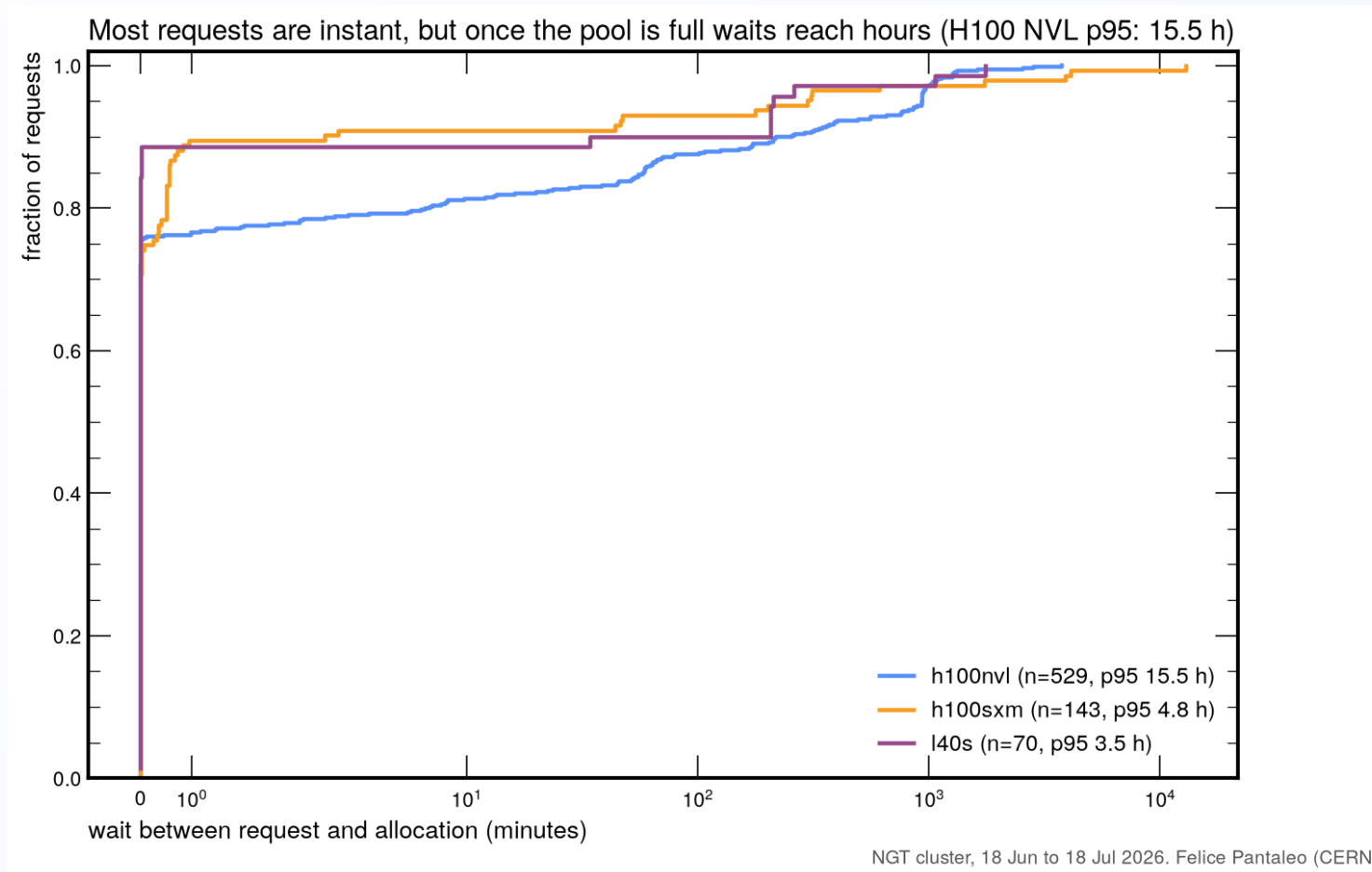
- All pools at their effective ceiling around the clock
- No diurnal breathing: allocation tracks ownership, not work
- Only the forced maintenance evictions of 6 to 8 July freed capacity



NGT cluster, 18 Jun to 18 Jul 2026. Felice Pantaleo (CERN)
NVL: MIG-enabled GPUs are not visible to DCGM, so the full-GPU ceiling sits below the 96-GPU capacity line. AMD from user-pod accounting (no DCGM).

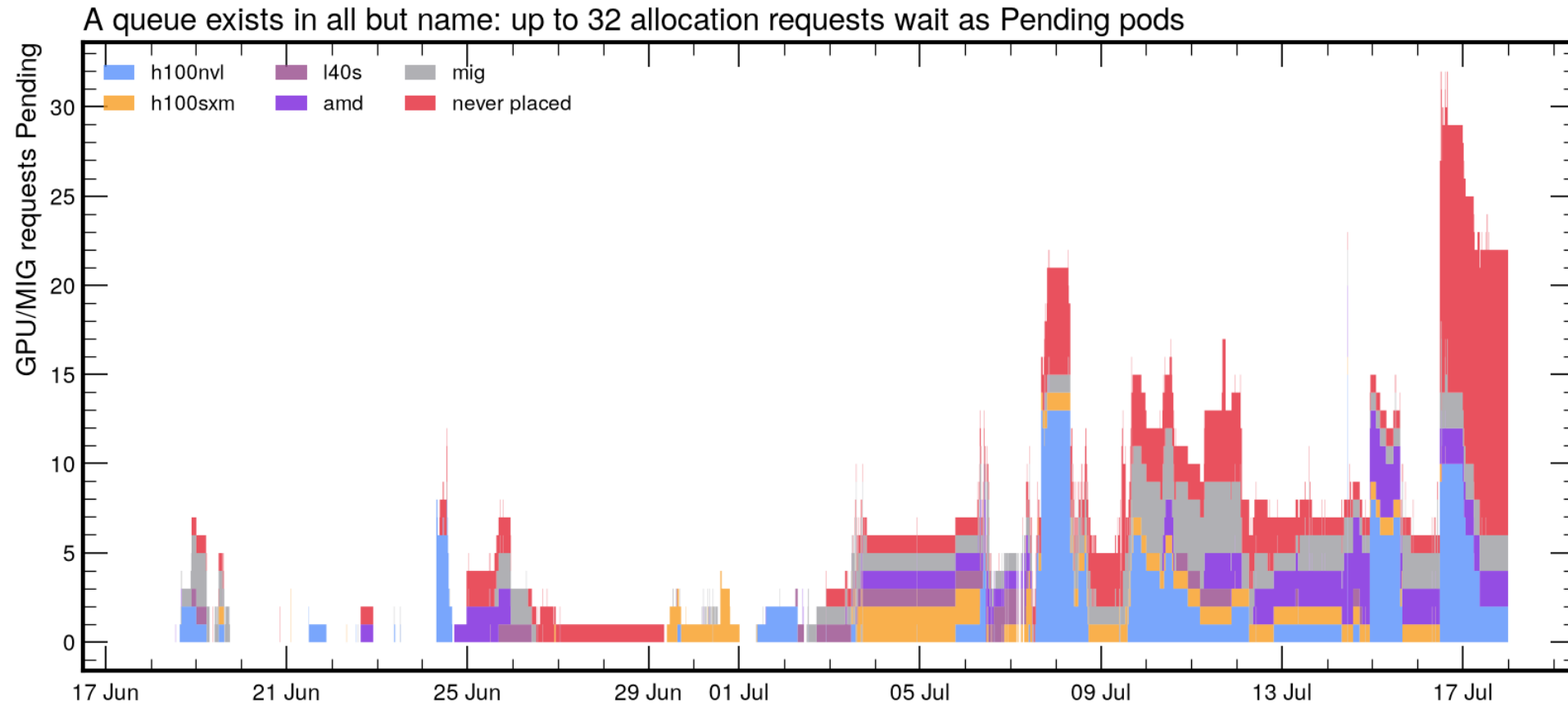
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When the pool is full, waits reach hours



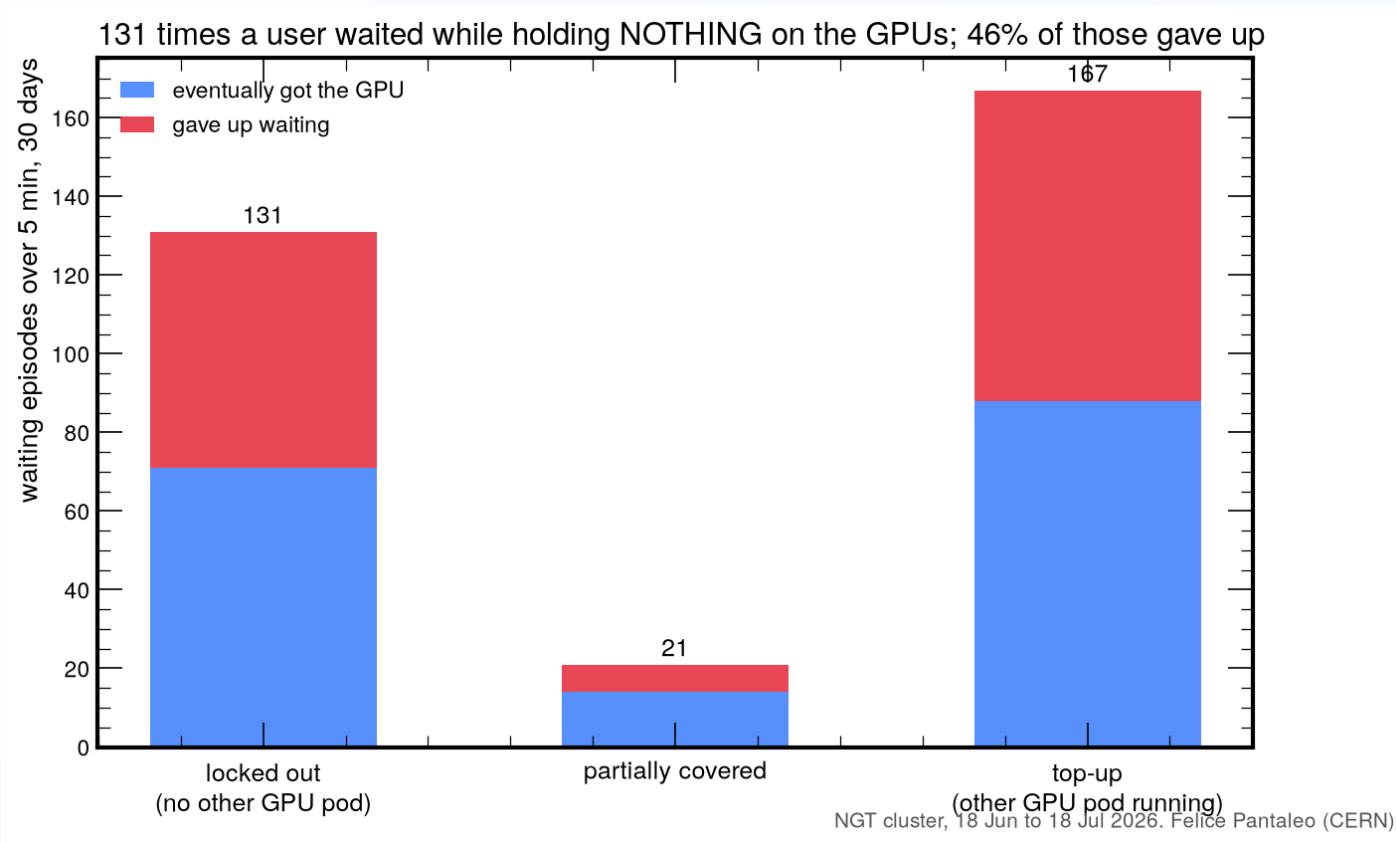
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A queue exists in all but name



NGT cluster, 18 Jun to 18 Jul 2026. Felice Pantaleo (CERN)

Who pays: 131 lockouts in a month

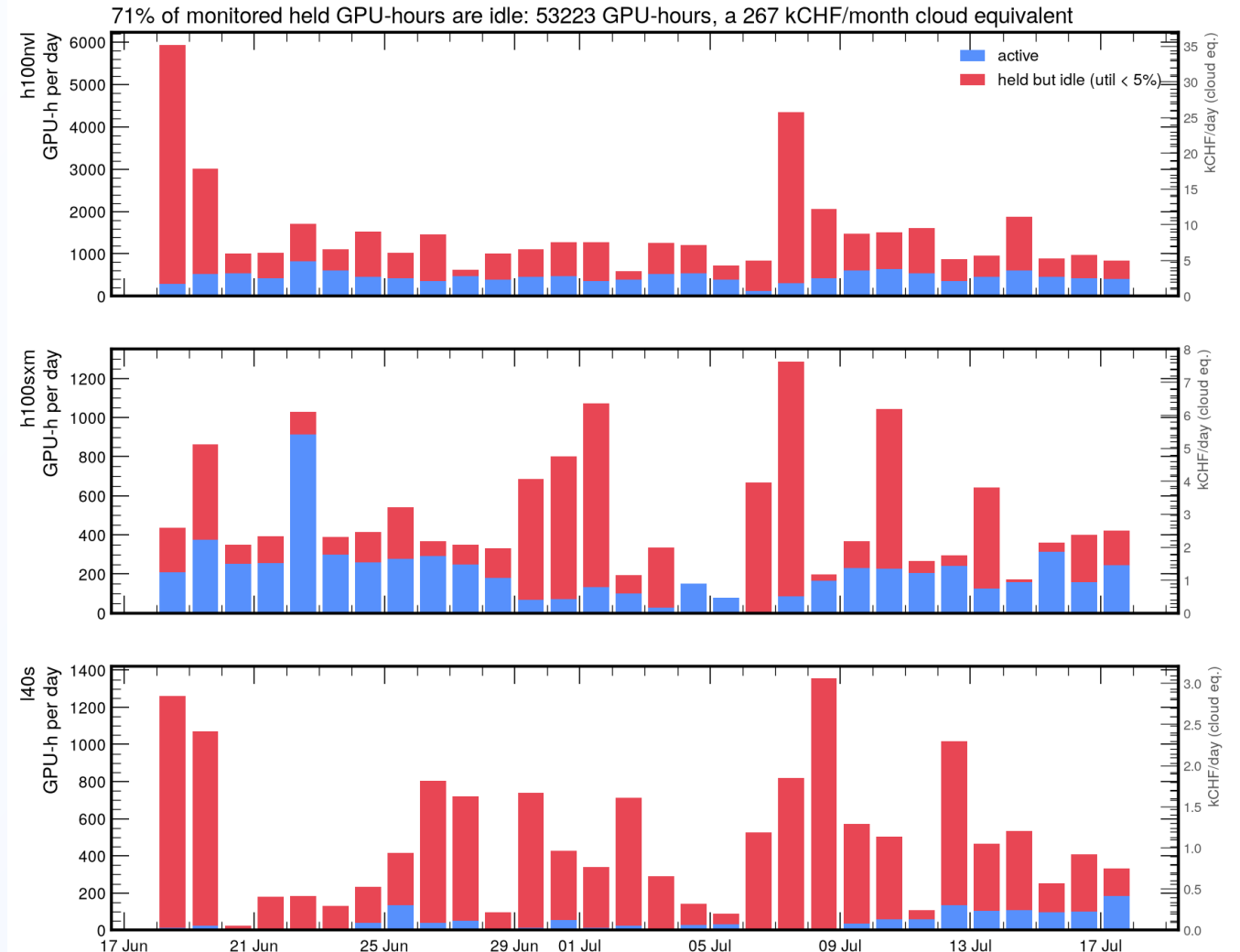


Over half of all queue pressure is top-up demand from users already running.

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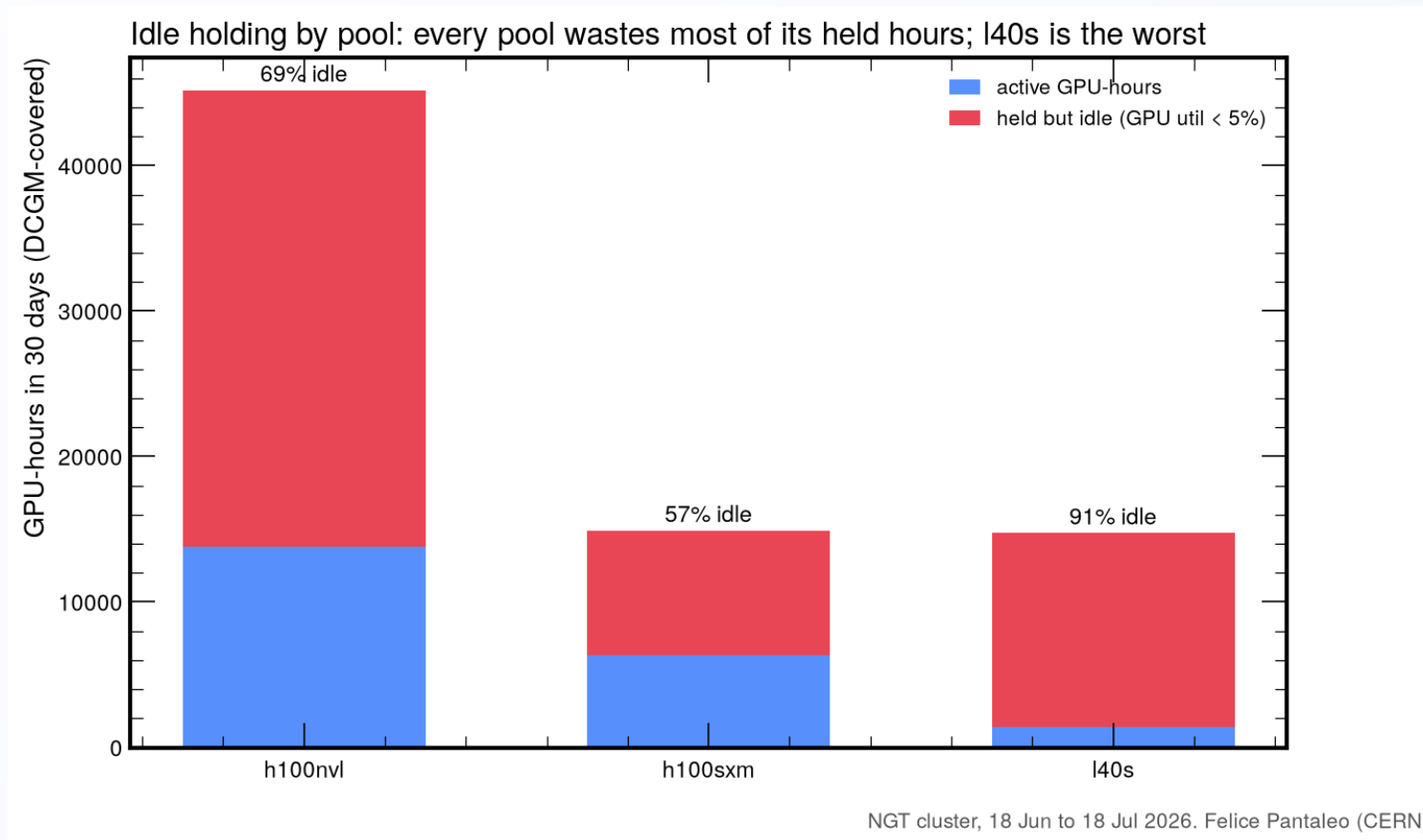
The cause: 71% of held GPU-hours are idle

- 53 000 GPU-hours parked in 30 days
- 267 kCHF/month cloud equivalent
- Right axes: kCHF/day per pool
- Continuous idle baseline on NVL and L40S; bursty parked episodes on SXM



DCGM utilization available for 500 full-GPU allocations; MIG slices and AMD not covered. Cloud rates as in the cost plot. NGT cluster, 18 Jun to 18 Jul 2026. Felice Pantaleo (CERN)

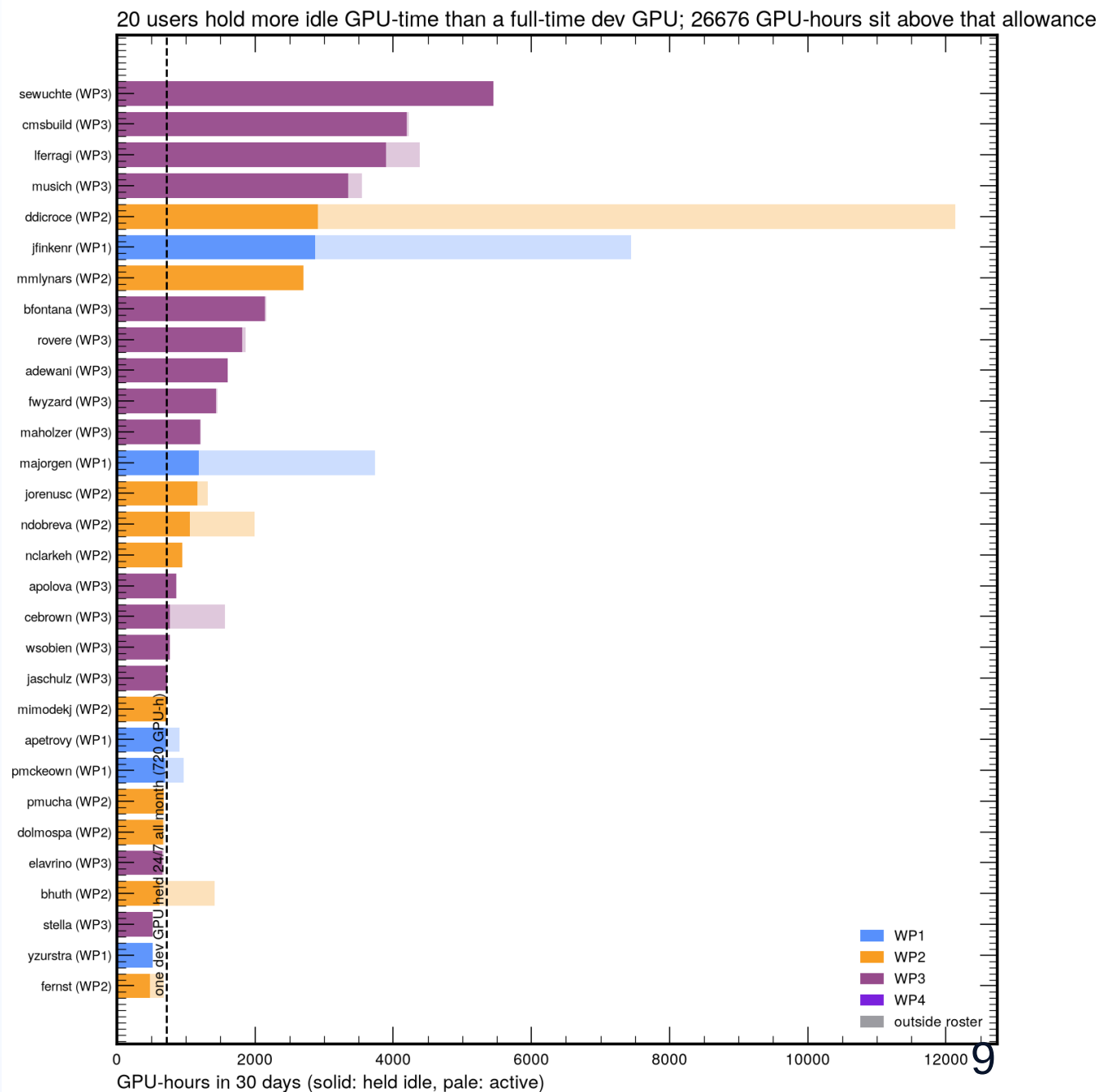
Every pool wastes most of its held hours



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Idleness, not usage, ranks the heavy holders

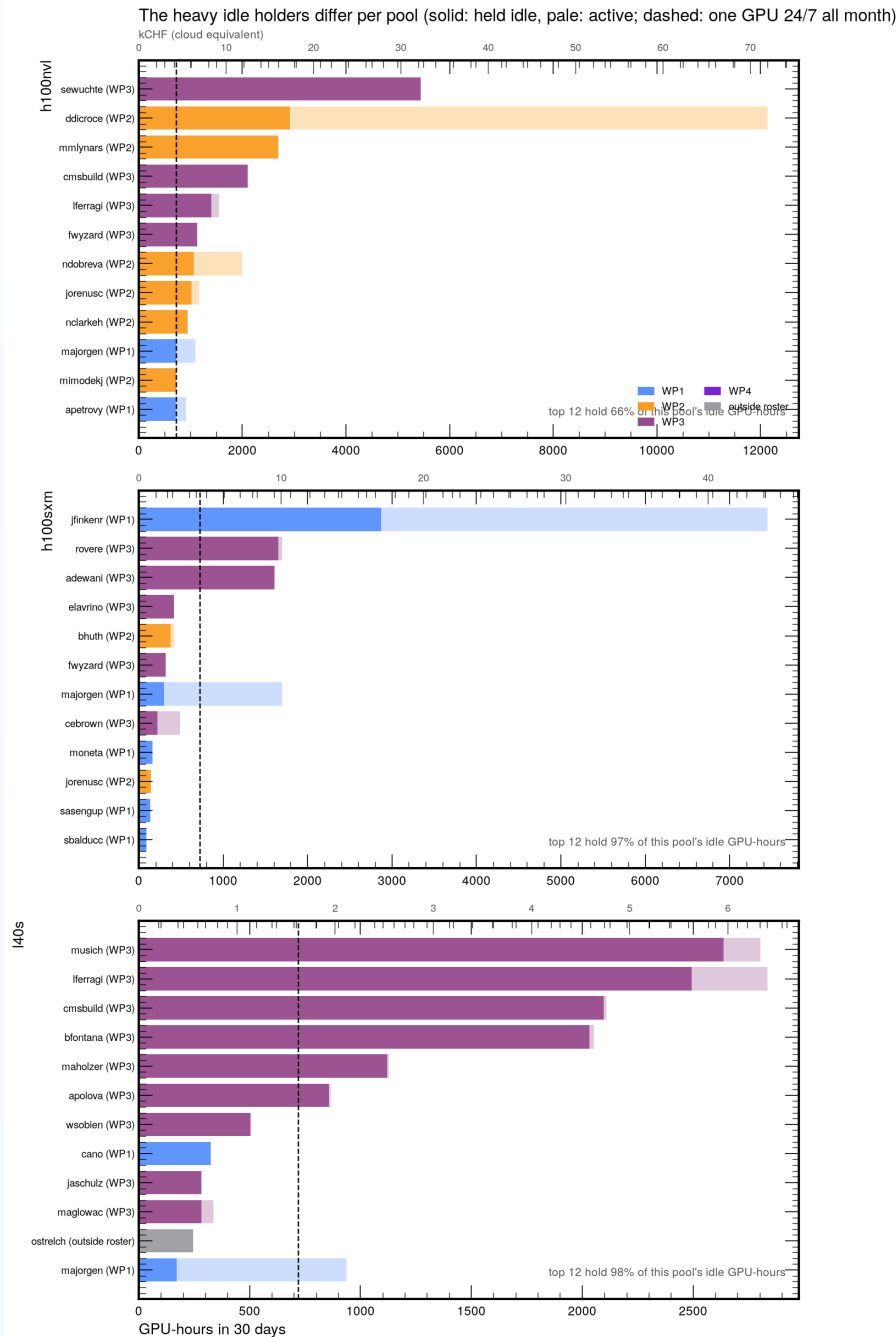
- Solid = held idle, pale = active
- Dashed line: one dev GPU held 24/7 all month (720 GPU-h), the legitimate P1 allowance
- **20 users** exceed it
- **26 700 GPU-hours** sit above it
- Greedy and productive are different people: the top consumer is almost all active



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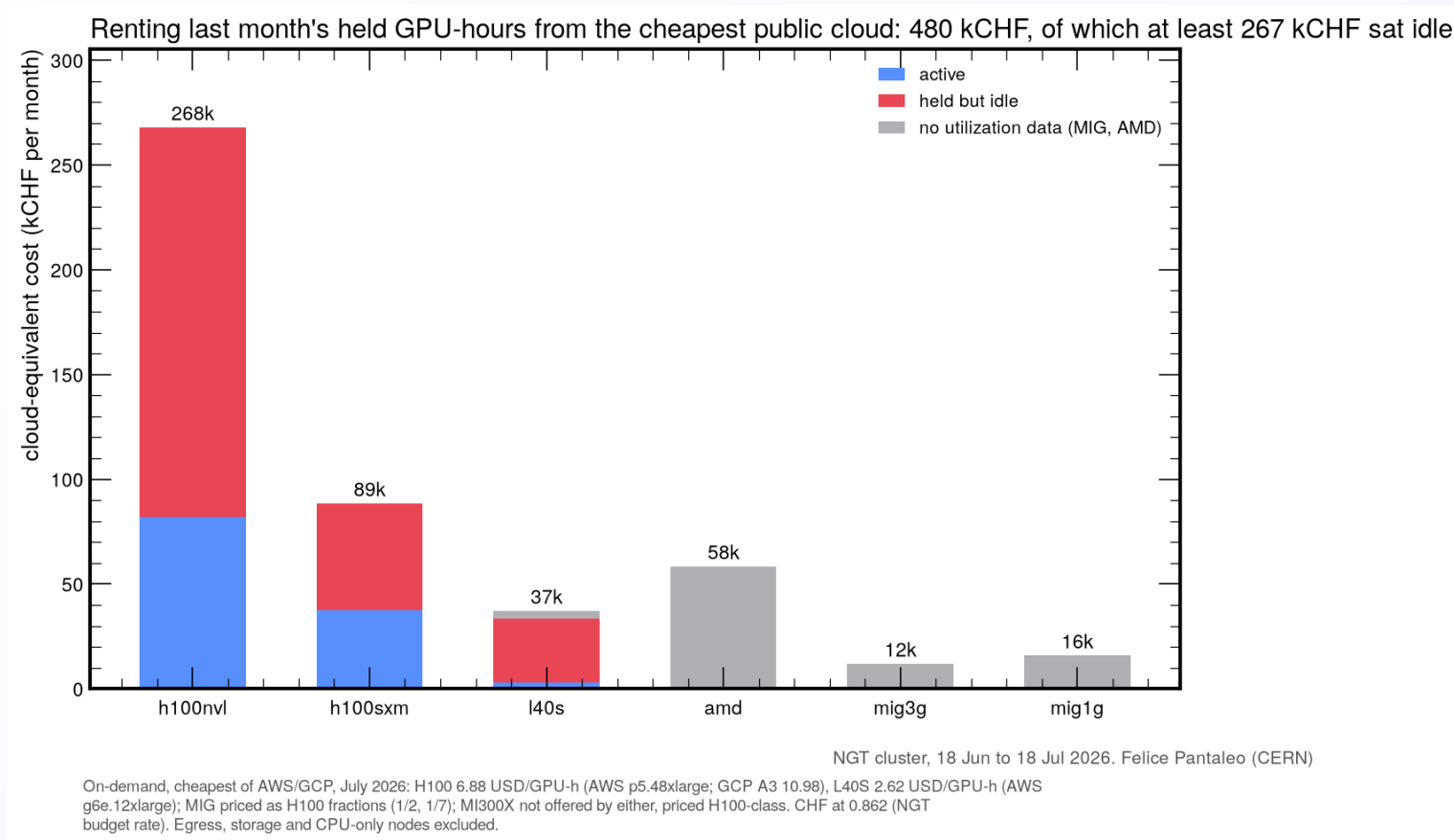
Where the greediness lives, per pool

- NVL: broad behavior, top 12 hold 66% of idle
- SXM: three users own the idle hours
- L40S: 12 users own 98% of the idle hours
- Automated reclaim fits NVL; L40S and SXM could be recovered with a handful of emails

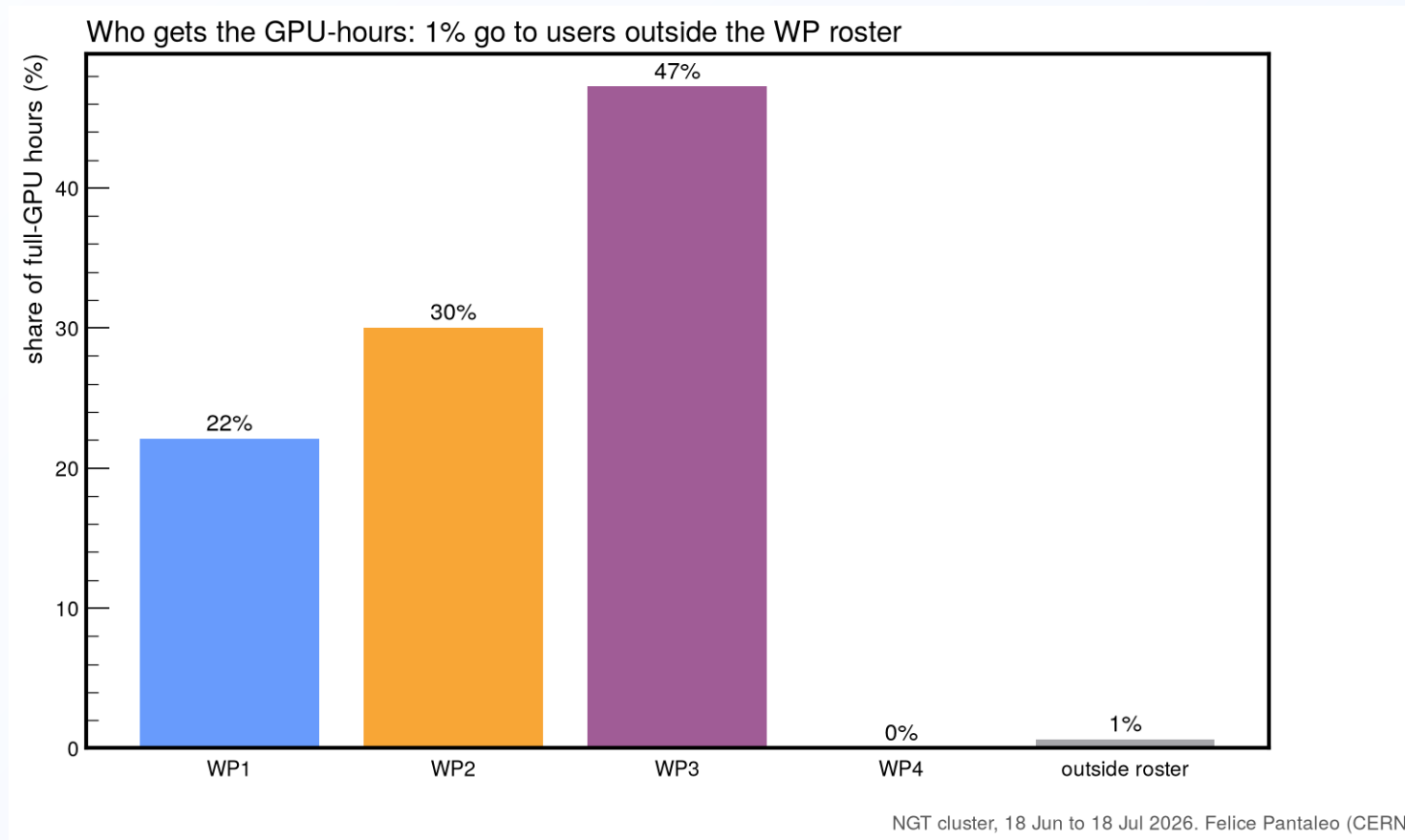


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What this would cost on a public cloud



Who gets the GPU-hours today



What a policy changes: the real month, replayed

Same 30 days, same requests, replayed through candidate schedulers:

	today (FCFS)	principles + idle reclaim
wait p95	557 min	4 min
requests satisfied	1393 / 1441	1430 / 1441
dev sessions within 15 min	89%	96%
multi-GPU holds over 24 h	96	0

Reclaim allocations idle longer than 30 min; guarantee one interactive GPU per member; cap multi-GPU holds at 24 h; enforce WP shares on live demand.

Proposal to the PMC

Reclaim idle holds

Reap GPU allocations idle for more than 30 min. Attacks 71% of the waste without touching active work.

Guarantee the dev GPU

One interactive GPU per member, always. Delivered in minutes in the replay; MIG carve spread over two nodes.

Cap and account

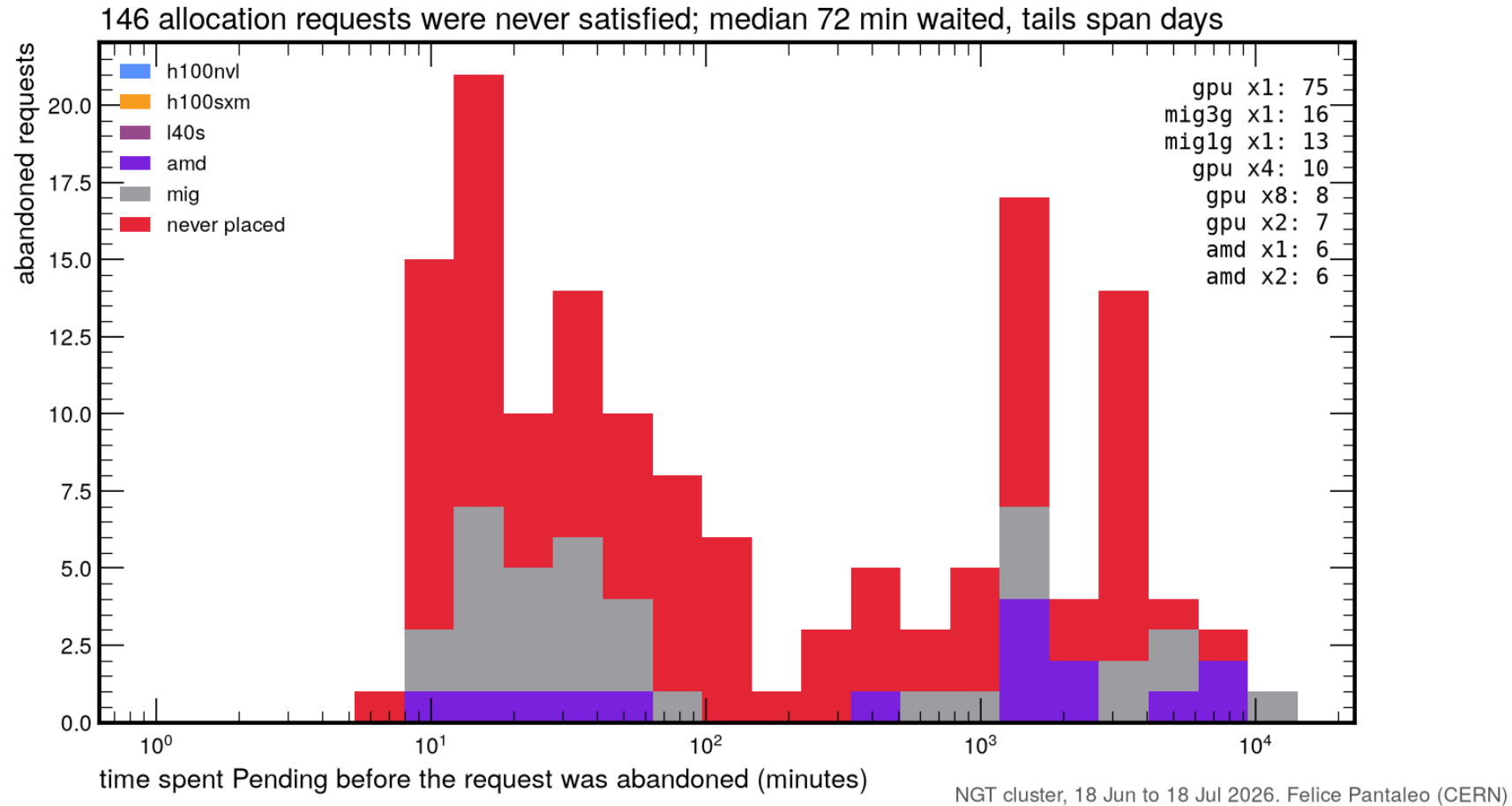
24 h cap on multi-GPU holds, GPU time charged to working packages with model correction factors.

All parameters simulated and tunable; policy simulator, data extraction and this evidence base are reproducible end to end.

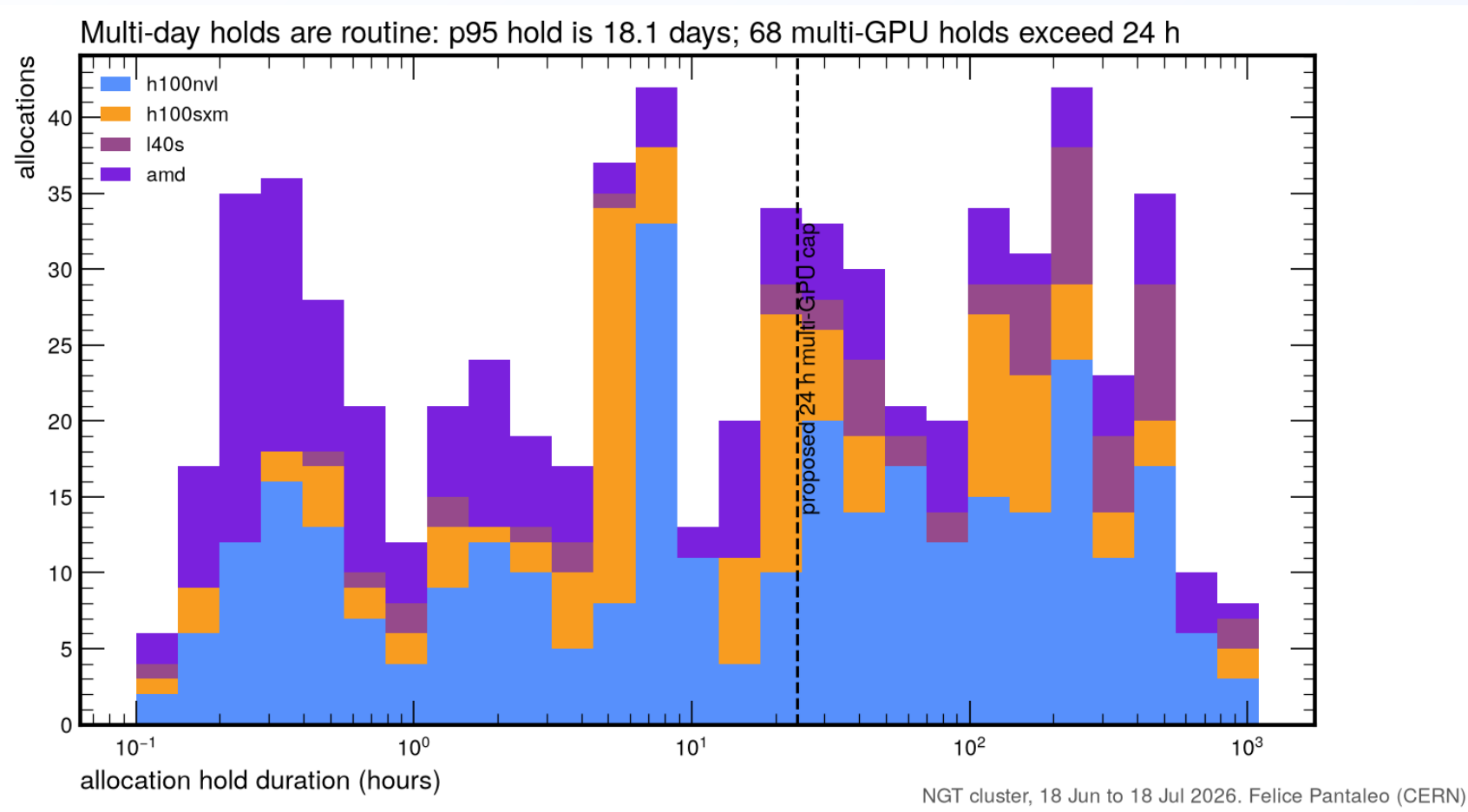
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Backup

Backup: unsatisfied requests



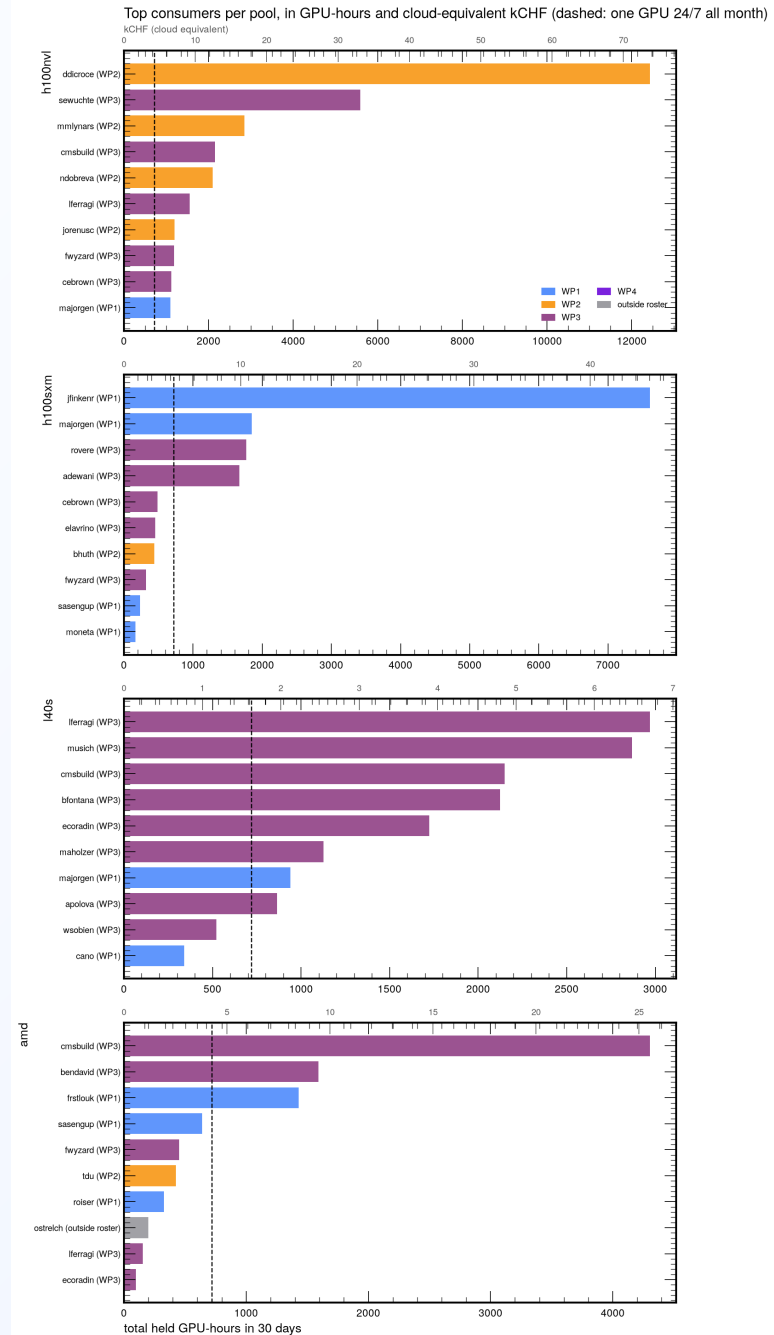
Backup: hold durations vs the 24 h cap



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Backup: top consumers per pool, in kCHF

- Secondary axis: cloud-equivalent kCHF, cheapest of AWS/GCP, July 2026
- Dashed: one GPU 24/7 all month
- 31 users held more than one GPU-month



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Backup: pool usage by WP, cordons, idle fractions

